

LedgerLens — Detailed Business Proposal

Accenture Innovation Challenge 2026 · Round 2 · Track 3, BusinessIntelligence.ai

A KPI intelligence-to-action engine. Root-cause analysis as a set intersection over a ledger of business changes, verified by negative controls, narrated by an AI investigator that proposes but never decides.

Repository: <https://github.com/omega-sus67/LedgerLens> · **307 passing tests** · runs end to end with no API key

Executive summary

Dashboards report *what* moved. Establishing *why* takes an analyst days across Slack, Jira, deploy logs and Zendesk — and the expensive failure is not the delay, it is acting on the wrong *why*.

LedgerLens takes an anomalous KPI, narrows it to the cohort that actually moved, intersects that cohort with every recorded change in the business, and then **tries to kill each candidate** with automatically generated negative controls. What survives is ranked, and every number on screen links to the SQL that produced it.

On the reference incident it diagnoses a **-\$416,144** renewals shortfall in **1.3 seconds**, and — the point of the whole design — it **rejects the plausible-but-wrong explanation** that a human under time pressure would likely have acted on.

Time to a defensible diagnosis	~3 analyst-days → 1.3 seconds
Numbers a reader can replay	22 distinct query_id s on one card
LLM cost per diagnosis	~\$0.0048 (Gemini Flash), \$0.0000 with the AI lane off
Cost of the wrong call it prevents	the -\$410k leak that stays open while you cut marketing

1. Problem framing

The observable problem

A dashboard reports that renewals in DACH fell 8%. It cannot say why. The question goes to an analyst, who spends the next three days reconstructing a narrative from Slack threads, deploy logs, support tickets and vendor emails — and by the time the answer arrives, the decision window has closed.

The expensive problem

The delay is not the costly part. Acting on the wrong "why" is.

In any given week, dozens of deliberate changes coincide with any metric movement. In our reference incident, marketing spend was cut in DACH **one day before** renewals broke. That correlation is real, it is temporally *more* convenient than the truth, and acting on it — restoring the campaign — costs money and fixes nothing. The actual cause was a payment-connector release that silently broke SEPA direct-debit renewals for enterprise customers in that region.

A tool that ranks by plausibility confidently recommends the wrong action here. That is the failure mode this system is built to eliminate.

Why this is hard, in the brief's own terms

Real-world complexity	Why naive approaches break
Multiple interacting drivers — price, mix, marketing, seasonality, competition	Correlation ranks the decoy first
Different grains, refresh cadences, historical coverage	A single global threshold makes a 98.2% rate KPI undetectable in principle
Sparse history for new products	Detection has nothing to fit a baseline against
Contradictory evidence, confidence calibration	A confident wrong answer is worse than no answer
Role-based security and sensitive dimensions	The same question must return different cuts to different readers
LLM economics — model choice, tokens, latency, cost	An LLM in the scoring loop is unbounded cost <i>and</i> unbounded risk

The claim we deliberately do not make

This does not prove causation, and it does not try to. With a single incident there is no population over which to estimate an effect, so any tool claiming "causal inference" here is overselling. What LedgerLens delivers is *ranked, auditable evidence plus the test that would settle it* — which is what an analyst actually needs in order to act.

2. Solution design

2.1 The organising principle

The LLM should not be treated as the source of quantitative truth. Teams should explicitly demonstrate when they use deterministic logic, SQL, statistics, business rules, traditional ML, causal inference, retrieval or LLMs — and why. — Round 2 brief, Track 3

This is the sentence the architecture is built around. Our answer is a **deterministic spine with probabilistic edges**:

Capability	Implemented with	Why that choice
Anomaly detection	Robust statistics — MAD z-score on rolling-median residual, STL when enough cycles exist	Outlier-resistant; needs no training data
Attribution / drill-down	Contribution analysis over a dimension lattice, top-down with a contribution floor	Bounds tests to the anomalous path, not the full cross-product
Multiple-testing control	Benjamini–Hochberg FDR	Labels, never gates
Context linkage	Change Ledger — typed events, each with a machine-readable <i>blast radius</i>	90% of enterprise linkage is already recorded. Foreign keys beat LLM entity extraction on cost, latency and error rate
Candidate generation	SQL set intersection of blast radius against the affected cohort	Sub-millisecond, explainable in one sentence
Ranking	Weighted rubric of five named components, each a query	A confidence number a judge cannot audit is a number that costs you points

Capability	Implemented with	Why that choice
Falsification	Automatically generated negative controls	The N=1-appropriate replacement for a causal-inference library's refuters
Learning	Beta–Bernoulli posterior counted from analyst verdicts	No model state to drift; deleting a verdict reverses it exactly
Proposing further checks	LLM , constrained to a template vocabulary	The rules cover the checks you anticipated; the LLM covers the ones you didn't
Naming untestable causes	LLM , in a separately labelled panel	The honest answer to "what if the cause was never recorded?"
Narration	LLM , behind a numbers guard	Prose is a rendering concern; numbers are not

2.2 The core primitive: blast radius

Every deliberate business change is recorded as a typed `ChangeEvent` carrying the exact slice of customers it *could* have touched:

```
deploy_sepa_v214    blast radius: {region: [DACH], payment_rail: [sepa]}
campaign_dach_cut   blast radius: {region: [DACH]}
```

"Is this change relevant to this anomaly?" becomes a **set intersection**, not a graph traversal or an embedding-similarity search. Deterministic, sub-millisecond, and explainable to a non-technical stakeholder in one sentence.

This is why the design needs no graph database, no vector store and no agent framework.

2.3 Scoring: five components, every one a query

$$\text{score} = 0.25 \cdot T + 0.30 \cdot C + 0.15 \cdot D + 0.25 \cdot N + 0.05 \cdot P$$

	Measures	Explicitly does not mean
T temporal	The change began shortly before the metric broke	Precedence is necessary for causation, never sufficient
C cohort match	Row-level Jaccard between blast radius and affected cohort	A wide radius scores badly even if it <i>was</i> the cause
D dose–response	Rank correlation of exposure against impact	Uninformative for this incident, and the card says so rather than manufacturing a number
N negative controls	Fraction of falsifiable predictions that survived	A passing control is a <i>failure to falsify</i> , not a confirmation
P learned prior	Beta–Bernoulli mean over past analyst verdicts	Weighted 0.05; sharpens a ranking, never decides one

2.4 The mechanism that matters: killing the decoy

For each candidate the engine constructs the cohorts its blast radius says should be **unaffected**, and the cohorts a rival mechanism says should **also** have moved, then checks whether reality agrees.

The marketing cut is a demand-side change to a *region*. If it caused the drop, DACH Mid-market and SMB customers should have fallen too. They did not — they came in **flat at -1.3%**. That control failure is decisive, and the decoy is **rejected outright at 0.322**, not merely outranked.

Candidate	T	C	D	N	Total	Outcome
deploy_sepa_v214	1.00	0.33	0.50	1.00	0.700	ranked #1
campaign_dach_cut	0.72	0.14	0.50	0.00	0.322	REJECTED

Being second is not good enough. A decoy still on the list is a decoy an executive might act on.

2.5 The AI investigator lane

Three LLM call sites, all **additive** to the deterministic spine:

1. **Proposed checks** — the model reads the anomaly, the ranked candidates and the controls already run, then proposes up to six further checks by filling a **fixed template vocabulary**. Our engine executes them in SQL. In the reference run it proposed a support-ticket check that **failed at +2766.7%** — a ticket spike that falsifies "this cohort was unaffected", which is evidence *for* the leading candidate.
2. **Unverifiable causes** — a separately labelled panel of explanations the connected data cannot test, each naming the feed that would settle it. The honest answer to *what if the real cause was never recorded?*
3. **Guarded narration** — persona-voiced prose, where every numeric token is checked against the verified payload. **One invented digit discards the narration** in favour of the deterministic template, and the page says so when it happens.

Nothing the model returns can change a rank. Enforced in code, not by convention: proposed checks are constructed with `decisive=False` — the field the N-scorer reads — and are never passed to it. A test asserts every score is byte-identical with the lane on and off.

Guards report what they caught. Hallucinated dimension values and attempts to escape the template vocabulary are rejected *before* becoming a query, and the count is displayed ("4 accepted, 2 rejected by validation"). A validator that never reports catching anything is indistinguishable from one that is not running.

2.6 Provider-agnostic by construction

One table of provider rows and one adapter class each. **Gemini 2.5 Flash by default** over hand-rolled REST; **Claude Sonnet** via the official SDK using forced tool-use. The two share no code path and no wire format, and the investigator cannot tell them apart.

```
LEDGERLENS_LLM_PROVIDER=anthropic # switch vendor, no source change
LEDGERLENS_LLM_MODEL=gemini-2.5-pro # switch model within a vendor
```

The `Provider` interface has **exactly one method**, and a test keeps it that way — a second method would necessarily describe one vendor's feature, and vendor-specific branching would follow.

2.7 Governance, security and honesty

- **KPI semantic contract** — definition, calculation SQL, drivers, thresholds, lineage, measured freshness and access rules, read by the engine and rendered in the UI. Not decorative metadata: the thresholds in force are the contract's.
- **Role-based entitlement** — enforced at exactly one chokepoint. Growth marketing loses the `payment_rail` cut, the card **names the policy** that withheld it, and the numbers legitimately differ because the focal cohort differs.

- **Abstention** — when no candidate clears the score floor, the system reports exactly that, lists which sources are and are not connected, and recommends widening ingestion. The failure direction is the design's saving property: **a too-wide blast radius fails its controls; a too-narrow one leaves nothing above the floor**. It degrades toward *I don't know*, never toward a confident wrong answer.
- **Provenance** — one function is the only path to the database. It hashes SQL plus parameters into a `query_id` and logs it, so every number on a card is replayable. There are exactly **two** deliberate exceptions, both facts about the *process* rather than the data — runtime telemetry and policy redactions — and both say so on screen rather than leaving a gap to be noticed.

3. Target users

Four personas render from **one computation**. Persona is accepted only by the narrator, which sits downstream of every query — so it *cannot* reach a query, and the evidence behind all four cards is identical by construction.

Persona	Channel	Depth	Decision rights	What they get
Revenue Analyst	workspace	full	all levers	Every control, every query id, the drill-down lattice
CFO	email digest	summary	hold_forecast	Dollars, forecast risk, an escalation — never an instruction to roll back a release
Payments On-Call	pager	operational	rollback_release , disable_flag	The event id, the blast radius, the rollback
Growth Marketing	workspace	summary	restore_campaign_budget	Their own KPI's real story — and a named policy where a cut is withheld

Decision rights are mechanical, not cosmetic. A persona that does not hold a lever is shown an *escalation*, never an instruction. A CFO is never told to roll back a release.

Recommendations follow the brief's chain exactly — **driver** → **controllable lever** → **action** → **expected impact** → **owner** → **confidence** → **monitoring plan** — plus a `basis` field carrying the `query_id` , so any recommendation opens onto the SQL underneath it.

4. Business case and impact

4.1 Stated assumptions

The brief invites reasonable assumptions, clearly stated. Ours, for a mid-market B2B SaaS business at roughly €50M ARR:

Assumption	Value	Basis
Material KPI movements needing investigation	~40 / year	Roughly weekly across 3–5 tracked KPIs
Analyst time per investigation	~3 days	The status quo the problem statement describes
Fully-loaded analyst cost	~\$460 / day	~\$120k total annual cost ÷ 260 days

Assumption	Value	Basis
Incidents where the first-guess cause is wrong	~1 in 4	The decoy pattern: a coincident change that is temporally more plausible

These are directional planning figures for sizing the opportunity, not measured results.

4.2 Where the value is

Analyst time recovered. 40 investigations × 3 days × \$460 ≈ **\$55k/year** of analyst capacity, redirected from evidence assembly to judgement. Real, but the smaller half.

Wrong-action avoidance — the larger half. In the reference incident the two costs of acting on the decoy compound:

- The **–\$410k** renewals leak stays open for however long the wrong fix is given to work.
- The campaign cut is reversed on a false premise, when the controls show it *did* cause a genuine ~31% new-logo drop — a real finding about a different metric.

One avoided misattribution per year on an incident of this size dominates the entire labour saving. This is why the negative-control layer, not the ranking, is the product.

Decision-window preservation. A diagnosis in **1.3 seconds** rather than three days means the finding lands while the incident is still live and the fix is still cheap.

4.3 What it costs to run

Diagnosis latency	1,293 ms cold; ~2.6× faster warm
Database work	89 registered queries executed, 41 served from cache
LLM cost per diagnosis	~ \$0.0048 on Gemini 2.5 Flash (3 calls, ~6k in / 1.2k out)
LLM cost with the lane disabled	\$0.0000 — the ranking path never calls a model
Same diagnosis on Claude Sonnet	~\$0.0240, no source change
Infrastructure	One DuckDB file. No graph DB, no vector DB, no agent framework, no orchestrator

At ~40 diagnoses/year the model spend is **under \$1/year**. Cost is not the constraint; trust is — which is why the engineering went into falsification and provenance rather than into a larger model.

4.4 Why this generalises

The only thing that changes per customer is the **connector mapping** from existing metadata to blast radius. A deploy already knows its rollout regions, a feature flag knows its targeting rules, a campaign knows its geo, a ticket knows its account and the account knows its segment. **The linkage is already recorded; it does not need to be inferred.**

5. Phased roadmap

Shipped — Round 2 prototype

Three KPIs across three sources with different grains; semantic contract; drill-down with contribution; five-component scoring; five negative-control rules; four personas; role entitlement; abstention; the AI investigator lane; the analyst feedback loop; runtime telemetry. **307 tests. All ten Minimum Prototype Expectations close.**

Phase 1 — Rigour (1–2 months)

Item	Why
<code>effect.py</code> — difference-in-differences with a bootstrap CI	Turn the observed shortfall into an interval; today no CI is shown because none is computed
Calendar-regressor baseline	Enables bidirectional detection; today only drops are flagged, because a quarter-end multiplier would flag every quarter close
<code>ambiguity.py</code> — the discriminating test	When two candidates sit within ϵ , name the one query that separates them
Bounded exploration pass	When nothing clears the floor, spend a capped query budget proposing new slices before abstaining

Phase 2 — Enterprise integration (3–6 months)

Real connectors (GitHub/GitLab, LaunchDarkly, Jira, Zendesk, campaign calendars, pricing tables) replacing synthetic fixtures — *the mapping is the only thing that changes*. Warehouse-native execution against Snowflake, Databricks or BigQuery, since the engine is already just SQL. SSO and per-user verdict attribution. Push delivery into Slack, email and PagerDuty on the channels each persona already declares.

Phase 3 — Proactive (6–12 months)

A watchtower that surfaces anomalies rather than waiting to be pointed; the LLM event normalizer for genuinely unstructured sources, **gated behind the confidence calibration work described in §6**; cross-KPI interaction detection; and a per-tenant learned prior once verdict volume makes it informative.

6. Key risks and mitigations

#	Risk	Mitigation	Status
1	The system names a confident wrong cause	Negative controls that try to <i>kill</i> each candidate; a decisive control failure rejects outright and cannot be outvoted	Shipped — the decoy is rejected on camera
2	The LLM hallucinates a number into the narrative	Numbers guard: every numeric token must appear in the verified payload; one invented digit discards the narration for the deterministic template	Shipped , and the discard is shown on screen
3	The LLM invents a dimension, metric or check	Validation against the dimension registry <i>before</i> execution; rejections counted and displayed	Shipped — "4 accepted, 2 rejected"
4	The LLM quietly influences the verdict	Proposed checks built with <code>decisive=False</code> , never passed to the scorer; test asserts scores identical with the lane on and off	Shipped
5	Blast radius metadata is wrong or missing	Failure <i>direction</i> is the mitigation: too wide fails its controls, too narrow leaves nothing above the floor. Degrades to "I don't know"	Shipped
6	Precision misread as proof	The card states outright that it ranks evidence and does not prove causation; the effect is labelled an observed shortfall, not a causal estimate	Shipped
7	Feedback poisons the ranking	P weighted 0.05 — a prior saturated with 200 confirmations still cannot lift a candidate over the floor or rescue one a control killed; both tested	Shipped
8	Sensitive dimensions leak across roles	Entitlement enforced at one chokepoint; the card names the policy rather than silently omitting	Shipped

#	Risk	Mitigation	Status
9	Vendor lock-in or model deprecation	One provider table, one adapter per vendor, one-method interface; two live adapters sharing no code path	Shipped
10	Vendor outage degrades silently	Every failure path returns empty with a recorded reason; the UI distinguishes "found nothing" from "vendor was down"	Shipped
11	LLM-extracted events outranking recorded ones	The event normalizer is deliberately not built — it is the one design in the original spec that puts a model on the ranking path	Deferred by decision, S5 Phase 3
12	Seasonality mistaken for an incident	Deseasonalized baseline; the card separates the -0.9% calendar component from the -7.5% that is real	Shipped
13	Verdicts unattributable to a person	No authentication in this build, so <code>verdict</code> records what was decided, not who	Named, not fixed — SSO in Phase 2

Appendix A — Minimum Prototype Expectations

#	Expectation	Where it lives
1	3–5 connected KPIs across 2–3 sources, different grains/cadences	<code>mrr_renewals</code> , <code>new_logo_bookings</code> , <code>payment_success_rate</code> (a ratio KPI)
2	Lightweight KPI/semantic contract	<code>contracts_decisions.md</code> — definitions, calculation SQL, drivers, thresholds, lineage, access
3	≥2 personas with different narratives/actions	Four — <code>persona_decisions.md</code>
4	One multi-factor KPI movement with known drivers	Seasonality + deploy + campaign decoy, decomposed on the card
5	One low-confidence scenario: clarification or abstention	Two — sparse KPI declines to detect; source-drop toggle makes refusal reachable (<code>abstention_decisions.md</code>)
6	One sparse-history / newly launched KPI	<code>sparse_kpi_decisions.md</code>
7	One role-based security scenario	<code>roles_decisions.md</code> — <code>fin.rail_detail</code>
8	Evidence: freshness, method, contribution, confidence, lineage	Contract panel + evidence chain, every step carrying a <code>query_id</code>
9	Clear LLM vs non-LLM breakdown	The 🕒 telemetry panel, on the page — <code>ai_decisions.md</code>
10	Runtime telemetry: latency, calls, tokens, cost	Same panel — three query counts, never merged

Ten of ten close.

Appendix B — Round 2 objectives

#	Objective	Status
1	Detects and prioritises material KPI movements	✅ MAD-z + practical-significance gate + BH labelling
2	Reconciles data and business context across heterogeneous sources	✅ Change Ledger; blast radius as dimension predicates
3	Ranks explanatory drivers using appropriate analytical methods	✅ Five-component rubric; every component a query

#	Objective	Status
4	Persona-specific narratives with traceable evidence	✓ Four personas, one computation, identical query ids
5	Communicates uncertainty; abstains when evidence is insufficient	✓ Score floor, abstention branch, connectivity report
6	Actions grounded in levers, constraints and decision rights	✓ The brief's full chain, plus a <code>query_id</code> basis
7	Mechanism to learn from analyst and business-user feedback	✓ Beta–Bernoulli prior from analyst verdicts — <code>learning_decisions.md</code>
8	Realistic security, cost, latency and scalability constraints	✓ Entitlement; 1.3 s; ~\$0.0048/diagnosis; one DuckDB file

Eight of eight.

Appendix C — What is deliberately not built

Naming gaps is the same discipline as abstaining on screen.

- `ledger/normalizer.py` — the LLM event normalizer. The one call site in our own design that is *not* additive: it multiplies a hypothesis score by an LLM-emitted confidence, putting a model on the ranking path. Schema and fields stay declared as the record of a deferred decision. (`ai_decisions.md` D2)
- `effect.py` — difference-in-differences with bootstrap CI. The impact figure is therefore reported as an *observed shortfall against a deseasonalized baseline*, with no confidence interval, because none is computed. We do not print an interval we did not calculate.
- `ambiguity.py` — the discriminating test between near-tied candidates.
- **Bidirectional detection** — v1 flags drops only; the rolling-median baseline does not model the quarter-end multiplier, so a two-sided detector would flag every quarter close. The correct fix is a calendar regressor, not a wider threshold.